MATH550 Numerical PDEs Homework #1

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Problem 1: Solve Stokes

First we solve for f: $\partial_{xx}u + \partial_{yy}u - \partial_x P = f$

$u = \sin(2\pi y)\sin(2\pi x)$
$\Rightarrow \partial_{xx}u = -4\pi^2 \sin(2\pi y)\sin(2\pi x)$
$\Rightarrow \partial_{yy}u = -4\pi^2 \sin(2\pi y)\sin(2\pi x)$

$P = \sin(2\pi y)\cos(2\pi x)$
$\Rightarrow \partial_x P = -2\pi \sin(2\pi y)\sin(2\pi x)$

$$\Rightarrow \boldsymbol{f = [2\pi(1-4\pi)]\sin(2\pi y)\sin(2\pi x)}$$

Next we solve for g: $\partial_{xx}v + \partial_{yy}v - \partial_y P = g$

$v = -3.5 + \cos(2\pi x)[\cos(2\pi y) - 1]]$
$\Rightarrow \partial_{xx}v = -4\pi^2 \cos(2\pi x)[\cos(2\pi y) - 1]]$
$\Rightarrow \partial_{yy}v = -4\pi^2 \cos(2\pi x)\cos(2\pi y)$

$P = \sin(2\pi y)\cos(2\pi x)$
$\Rightarrow \partial_y P = 2\pi \cos(2\pi y)\cos(2\pi x)$

$$\Rightarrow \boldsymbol{g = [2\pi(1-4\pi)]\cos(2\pi y)\cos(2\pi x) + 4\pi^2 \cos(2\pi x)}$$

$h = 0$ because $\partial_x u + \partial_y v = 0$

1. Solution plot of $|\boldsymbol{u}|$:

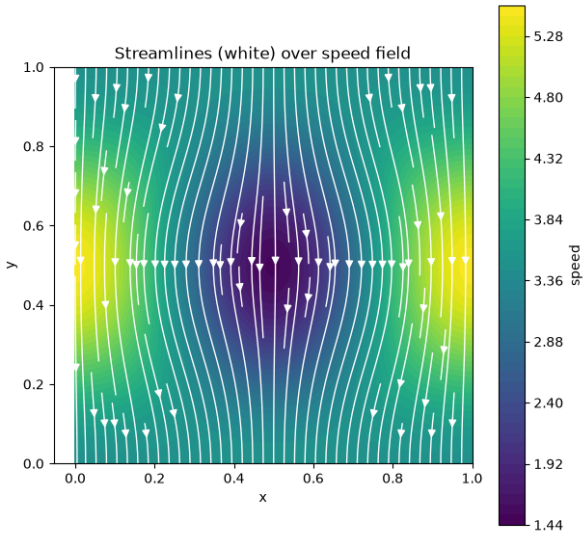


Figure 1: Chart of exact vs. numerical solution

2. Error plot of $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{p}$:

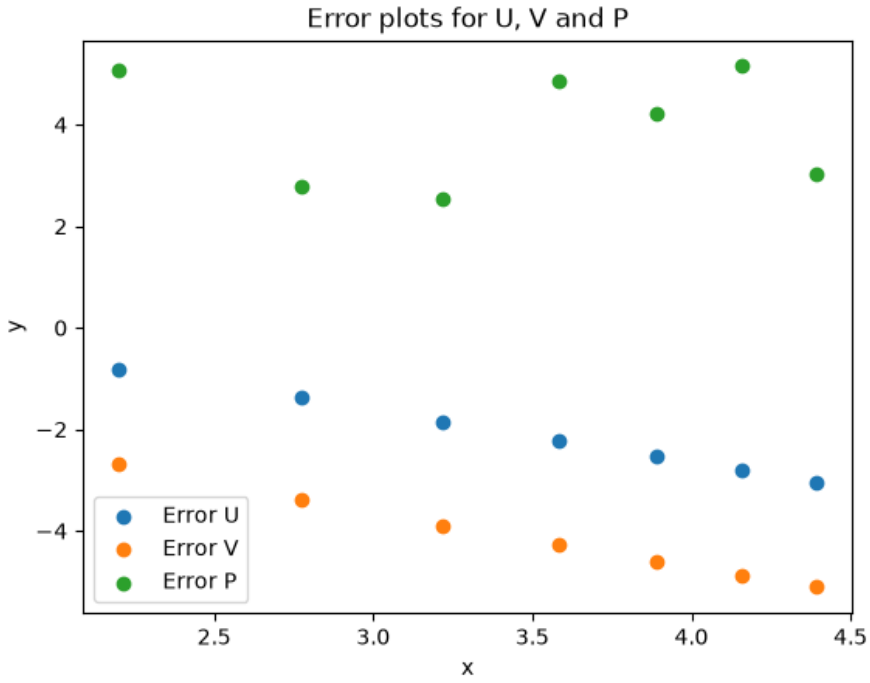


Figure 2: Chart of exact vs. numerical solution

It is odd that the error for P does not follow the same trend as the error for U or V.

Use of AI

I used deepseek to generate plots. I attempted to use deepseek to debug the code, but was ultimately unsuccessful with that approach. It turns out that I was calculating $-D_y$ instead of D_y. I found this error, not AI.